

Impact Benefits

Expanded stakeholder-impact and measurable-benefit report for Slide 5. The original slide maps benefits to MOIL Limited, Ministry of Steel, mine planners/engineers and downstream steel manufacturers. [filecite]turn1file0[L33-L36]

1. MOIL Limited

MOIL is the most direct operational beneficiary because the platform can connect exploration, reserve estimation and production planning. Its FY25 report records 121.97 Mt reserves/resources and 107,530 m of exploratory drilling, showing the scale of the geological information problem. [cite]turn0search48

The platform can create a common view of where additional exploration evidence is needed, where uncertainty is high, and how current operational constraints affect planned production.

2. Ministry of Steel

The national-level value is improved visibility of manganese supply risk. MOIL's FY25 MD&A reports 3.75 Mt Indian manganese ore production and about 6.58 Mt imports, illustrating the scale of domestic supply versus import dependence. [cite]turn1search24

NAKSHATRA-X should therefore be positioned as a decision-support and resilience layer for the mineral-to-steel value chain, not as a replacement for mining governance.

3. Mine planners and engineers

The most immediate operational benefit is early warning. A useful alert should answer four questions: what changed, why it matters, how much output is at risk, and what feasible intervention exists?

For example, a rainfall anomaly can become a production-risk scenario when combined with haulage accessibility, pumping status and historical disruption behaviour.

4. Downstream steel manufacturers

More predictable ore supply can support procurement and production planning. However, this should be stated as an intended downstream benefit unless the team has direct customer/plant data showing the effect.

5. Environmental and safety value

Satellite rainfall and SAR can improve situational awareness during flooding or difficult weather. ESA specifically documents Sentinel-1's ability to image through cloud/rain and its usefulness for flood and ground-monitoring applications. [cite]turn0search3[turn0search8]

The system should prioritize warning and decision support. Any automated pump or SCADA action must remain governed by validated control logic and site safety procedures.

6. Quantifiable impact framework

Reserve: increase in validated target hit-rate; reduction in uncertainty; reserve/resource added per exploration metre.

Production: reduction in planned-vs-actual shortfall; tonnes recovered after intervention; equipment utilization.

Risk: warning lead time; false-alert rate; disruptions detected before operational impact.

Grade: blend deviation; off-spec tonnes; assay-prediction error.

Economic: avoided cost + recovered contribution – implementation and operating cost.

7. National-scale story

IBM's Indian Minerals Yearbook provides an official statistical reference series for India's mineral industry, while its manganese tables identify major producers and mining locations. [cite]turn1search1[turn1search26]

The strongest impact claim is therefore not a single large rupee number. It is a measurable chain from better evidence → better decision → measurable operational delta → verified economic value.

8. Closing impact statement

“NAKSHATRA-X turns space and geological data into operational intelligence: helping mines understand where uncertainty is highest, when production risk is rising, and which feasible response should be evaluated next.”

Evidence and exact source paths

ID	Source	Exact resource / URL	Use in NAKSHATRA-X
S1	MOIL 63rd Annual Report 2024-25	https://moil.nic.in/userfiles/Annual_Report_2024_25.pdf	Reserves/resources, exploration drilling, production and strategic target.
S2	MOIL MD&A; / Annual Report copy	https://moil.nic.in/userfiles/file/InvRel/Financials/Annual_Report_2024-25.pdf	FY25 manganese production and import context.
S3	MOSDAC — GSMap ISRO Rain	https://www.mosdac.gov.in/gsmmap-isro-rain	Exact rainfall product; hourly, 0.1° and March 2000 onward.
S4	GSI Reports database	https://gsi.gov.in/gsiReport/	Official geological/exploration report search.
S5	GSI Mineral Hunt data portal	https://hackathon.gsi.gov.in/login	Direct geology, exploration and remote-sensing data routes.
S6	GSI geology data	https://nis.bisag-n.gov.in/download/Geology/	Direct geology data directory.
S7	GSI exploration reports/data	https://nis.bisag-n.gov.in/download/Exploration_data_reports/	Direct exploration data/report directory.
S8	GSI remote sensing ASTER	https://nis.bisag-n.gov.in/download/Remotesensing_ASTER/	Direct remote-sensing directory.
S9	GSI/OGD manganese deposit dataset	https://tn.data.gov.in/catalog/location-manganese-ore-deposits-in-dia-and-its-salient-features	Specific manganese-deposit location/geology dataset.
S10	Indian Minerals Yearbook portal	https://ibm.gov.in/IBMPortal/pages/Indian_Minerals_Yearbook	Official IBM source for manganese statistics.
S11	Indian Minerals Yearbook 2024	https://ibm.gov.in/writereaddata/files/177426215469c1178a48453IMYB_2024_EBookFinal.pdf	Producer/location and manganese industry data.
S12	ESA — Sentinel-1 seeing through clouds	https://www.esa.int/ESA_Multimedia/Videos/2014/03/Sentinel-1_seeing_through_clouds	SAR capability through cloud/rain and day/night.
S13	ESA — Introducing Sentinel-1	https://www.esa.int/Applications/Observing_the_Earth/Copernicus/Sentinel-1/Introducing_Sentinel-1	C-band SAR, all-weather/day-night mission context.